

Process or Transmit?

A value-aware onboard decision engine for resource-constrained satellites

Every scarce resource gets a price; every data item picks its best net-value option. On Sentinel-2A's real orbit: most science value in 5 of 6 scenarios against four baselines, with up to 22 % less energy.

QR code
repo + live
demo
(placeholder)

Team name · Affiliation · IASTAM 2026 · Problem 1

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scenarios where ours captures the most value, winning 18–20 of 20 real days in each

+2–20 %

more value than the best of four baselines (up to +91 % vs process-all)

–22 %

energy for processing + downlink vs the rule set, nominal mission

28 % vs 10 %

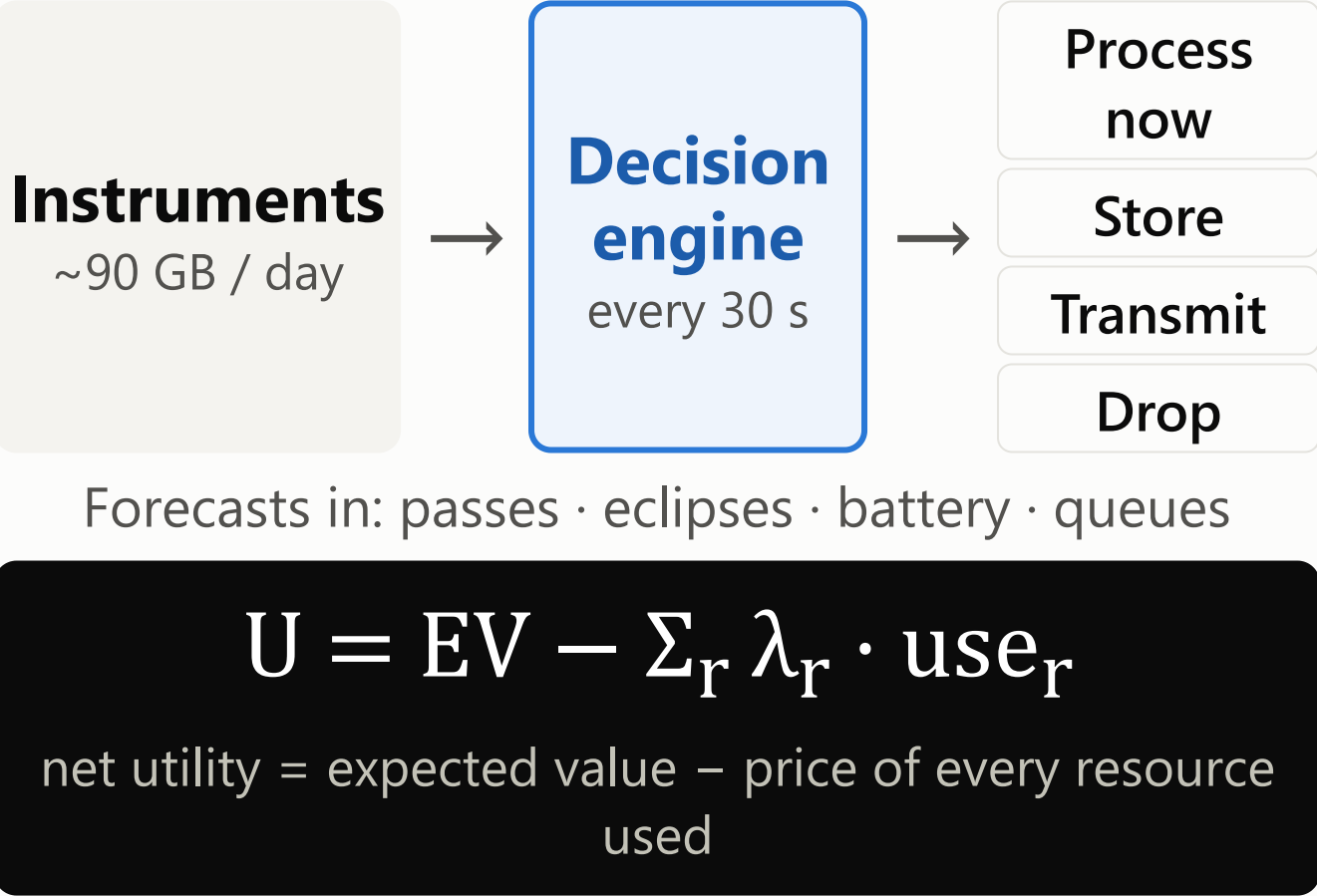
mean lowest battery when energy-starved: ours vs process-all (held at the load-shedding floor)

1 The problem

A small Earth-observation satellite generates **~90 GB/day** but can downlink only **~21 GB/day**. For every item: **process now, store, transmit**, or drop?

- Real orbit: Sentinel-2A TLE (786 km SSO), propagated with SGP4 (skyfield)
- Battery charges only in sunlight: 34-min eclipse every 101-min orbit
- 1 GPU + 4 CPU cores, 8 GB RAM, 32 GB storage
- ~4.4 real passes/day over Toulouse, up to 10 min, 80 Mbit/s
- Value decays with latency: an alert loses half its value per hour

2 Our idea



- Value** each option (raw vs processed): priority × P(useful) × timeliness.
- Price** downlink, energy, GPU, CPU, storage: prices rise until the plan fits forecast passes, sunlight and disk.
- Choose** the best option per item: PROCESS / RAW / NONE.
- Act**: start jobs, order the downlink, guard storage. Re-planned every 30 s.

Take-aways

- Prices turn five coupled constraints into one comparable number per option.
- Biggest gain when the downlink is scarce: +20 % value vs the best baseline. When memory is the bottleneck, simply processing everything is best.
- Every choice is logged with its utilities and prices, so operators can audit it.

3 How we evaluated

Simulator of power, compute, storage and passes on 20 real days of Sentinel-2A's orbit. **6 scenarios × 5 strategies × 20 days = 600 simulated days.**

Scenario	Stress: 2/3 of what processing all needs
Nominal	~90 vs ~21 GB/day
Energy-starved	44.8 W solar, 60 Wh battery
Downlink-starved	20° mask, 40 Mbit/s
Storage-tight	2.85 GB mass memory
Compute-starved	4.7× slower accelerator
Event surge	5× urgent events

- Value-aware (ours)
- Priority rules
- Process-all
- Bent-pipe
- Bandwidth-aware rules

Metrics for the 5 criteria: value captured · completed tasks · energy & lowest battery · latency (alerts) · GPU, downlink, storage use.

4 Explainable decisions

Event candidate → **PROCESS NOW**
U_proc 10.66 > U_raw 7.54; ~1 MB product vs 33 MB raw

Hyperspectral cube → **TRANSMIT RAW**
U_raw 4.47 >= U_proc 2.74 (spare downlink: raw keeps more science)

Optical image → **DROP**
storage guard, lowest value per MB onboard [storage saturated]

Real log lines, seed 7 (nominal and storage-tight runs).

6 Limits & next steps

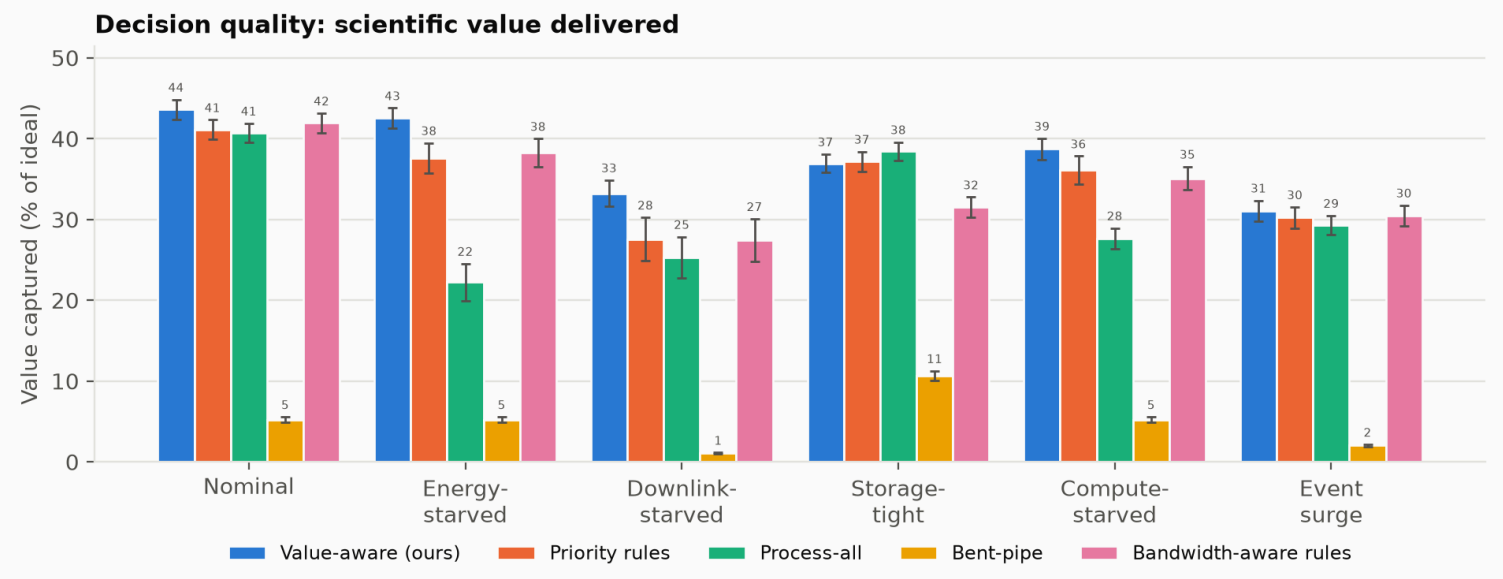
Limits

- Real orbit and passes; illustrative satellite resources and workload
- One ground station, simple power model
- When memory is the bottleneck, process-all wins (–1.5 pts)

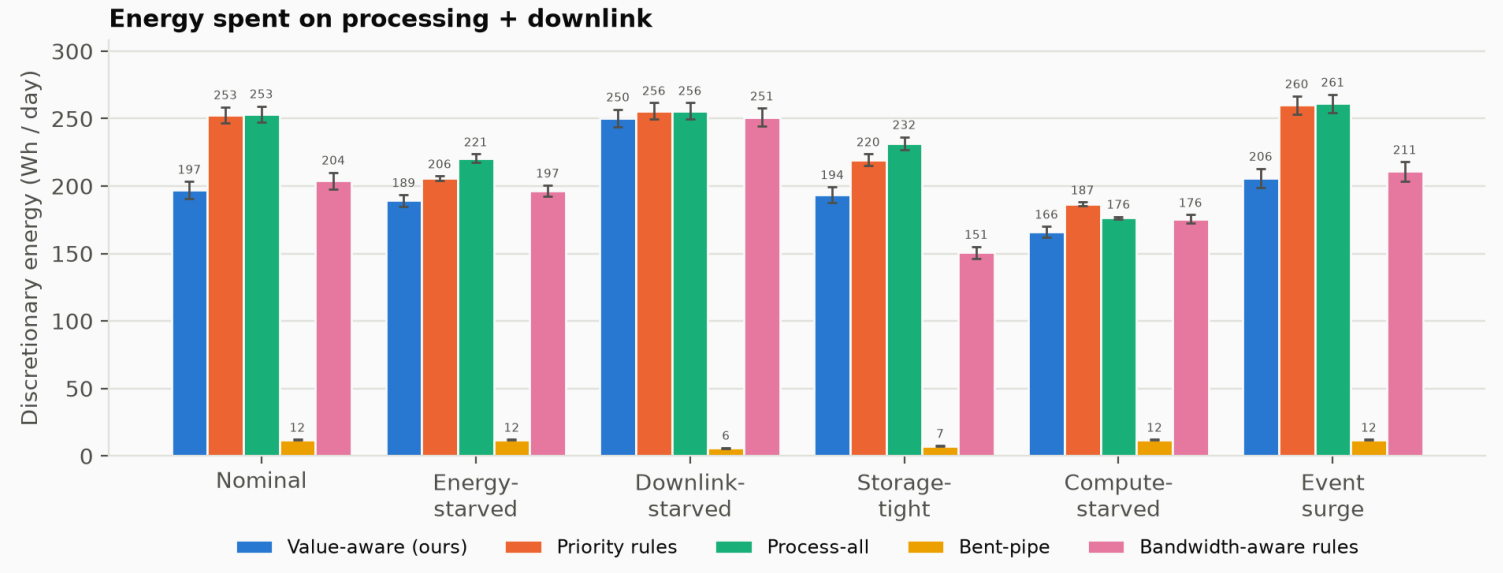
Next

- Flight compute costs on target hardware
- Hardware-in-the-loop tests
- Learned usefulness, real mission data

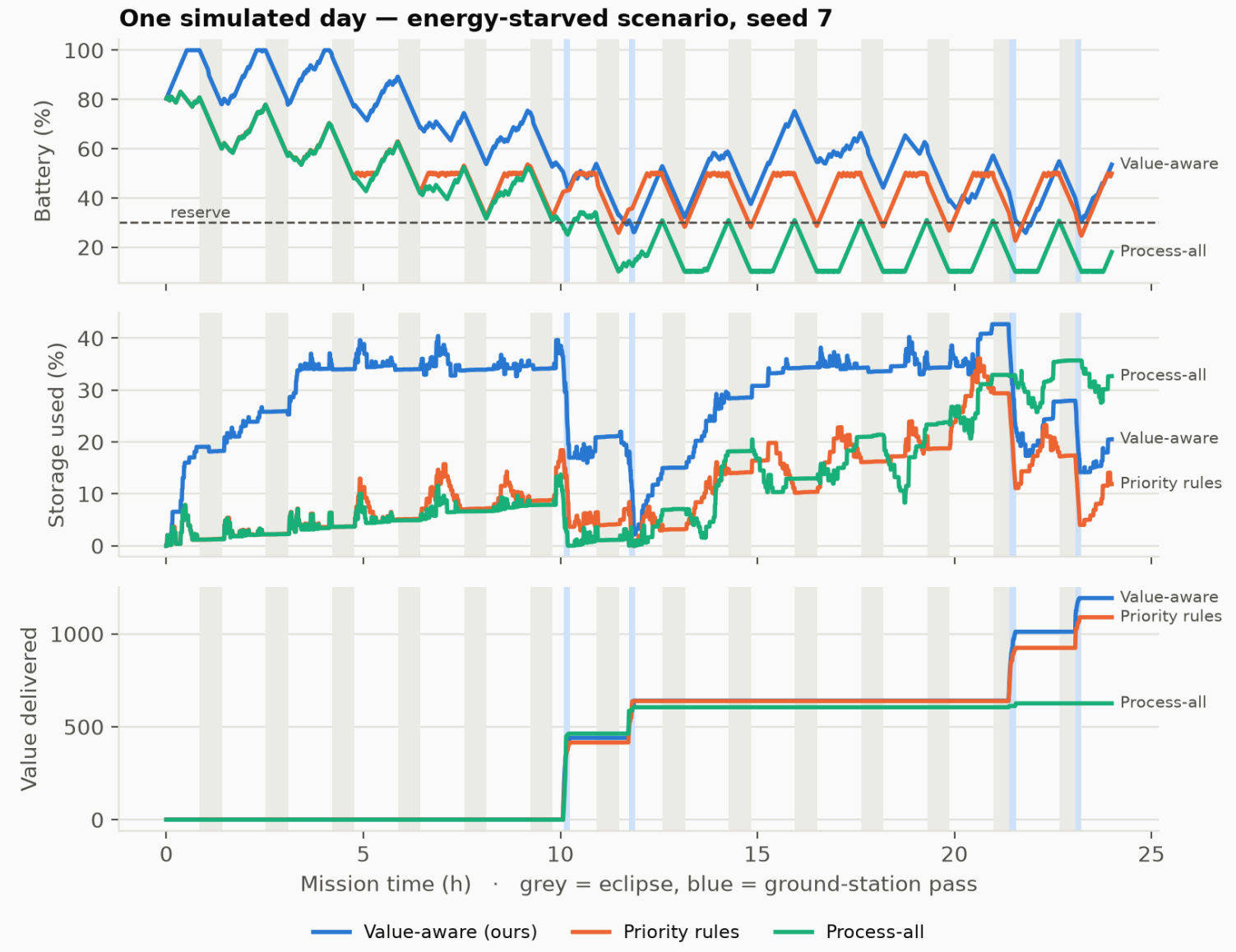
5 Results



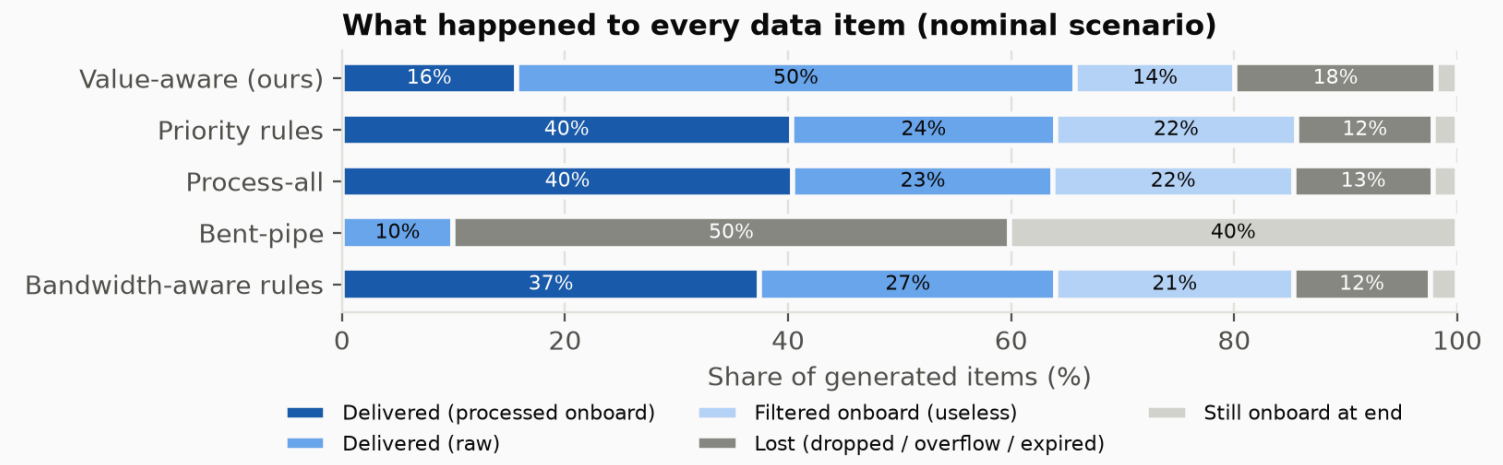
Most value in 5 of 6 scenarios; process-all is best when storage is tightest (mean of 20 real days, 95 % CI).



2–22 % less energy than the rule set and process-all.



Energy-starved day (real passes and eclipses of 3 Oct 2026): process-all is held at the load-shedding floor; ours keeps a margin and delivers the most value.



Nominal: ours delivers the most items (65.0 vs 63.7 % on average) with none expired; completion is lower (79 vs 85 %) because it drops undeliverable items early and sends raw when bandwidth allows.